



## E-Prescribe Decision Guide

*Companion to the [E-Prescribe \(eRx\)](#) docs.*

E-prescribing in Medplum runs through an integration with an e-prescribing vendor connected to the SureScripts network. This space is less about the FHIR data model than a sequence of **integration and enrollment decisions**: how much prescribing UI you build, whether you send controlled substances, and how prescribers get enrolled and verified. This guide moves from those high-level decisions down to feature detail.

### Section 1: Use Case & Participants

*Two of these — what you prescribe (1.1) and how much you build (1.2) — cascade into most of Section 3. Land them first.*

#### 1.1 What are you prescribing?

- Non-controlled medications only
- Controlled substances too (Schedule II–V)
- Not sure / depends on specialty

*Why: controlled substances require **EPCS**, which adds a per-prescriber enrollment path (DEA, identity proofing, two-factor auth). If you never prescribe them, skip the entire Controlled Substances lane (3.4) and enrollment shrinks substantially.*

#### 1.2 How much of the prescribing experience do you want to build?

- **Hosted iframe** — embed the vendor's prescribing UI. Minimal build; drug search, dosing, pharmacy, and send all happen inside the vendor's screens.

- **Integrated (API-driven)** — build your own prescribing UI on Medplum's FHIR operations and bots; only the final, regulation-required **review-and-send** step stays a hosted widget.
- Not sure

*Why: the core build-vs-UX trade-off. The iframe is fastest to live but UX is fixed and you generally can't prescribe from inside your own chart.*

*Integrated lets you own the workflow and prescribe in context, at the cost of building it. See 3.1.*

### 1.3 New build or replacing an existing e-Rx system?

- New — no e-prescribing today
- Replacing another EHR or e-Rx vendor
- Adding alongside an existing clinical system

*Why: replacing an existing system means a **Change of Vendor** on SureScripts so refills and history follow prescribers over — a distinct migration workstream (3.11).*

### 1.4 Who is involved in prescribing?

- Prescribers (sign and send)
- Prescribing agents / proxies (stage but can't send)
- Administrators (enroll prescribers, manage practices, configure favorites)
- Any offshore or non-US staff using the prescribing UI?

*Why: prescriber vs. agent vs. admin map to distinct enrollment roles (3.3), and only fully-enrolled prescribers can transmit. Non-prescribing roles don't need an NPI. Offshore and non-US staff matter here because e-prescribing is only available to professionals **authorized to prescribe in the United States** — a SureScripts-network constraint, not a Medplum one — so non-US staff can only hold non-prescribing roles, and even those must be enrolled with the vendor to call pharmacy or medication-history operations (3.3). Confirm your vendor's policy on non-US users before designing an offshore-staffing model around the prescribing UI.*

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## Section 2: Feature Scoping

For each row, mark Yes / No / Nice-to-have / Not sure. The § column points to the deep dive.

| #  | Feature                                              | §   | Yes | No | Nice-to-have | Not sure |
|----|------------------------------------------------------|-----|-----|----|--------------|----------|
| 1  | Integration model (hosted iframe vs. integrated/API) | 3.1 |     |    |              |          |
| 2  | Prescriber enrollment & identity proofing            | 3.2 |     |    |              |          |
| 3  | Provider & practice identity data (NPI, DEA, roles)  | 3.3 |     |    |              |          |
| 4  | Non-prescriber staff access (pharmacy / history ops) | 3.3 |     |    |              |          |
| 5  | Controlled substances / EPCS                         | 3.4 |     |    |              |          |
| 6  | Formulary/benefit checks & prior authorization       | 3.5 |     |    |              |          |
| 7  | Refills, reorders & discontinuation                  | 3.6 |     |    |              |          |
| 8  | Clinic favorites / order sets                        | 3.7 |     |    |              |          |
| 9  | Pediatric prescribing                                | 3.7 |     |    |              |          |
| 10 | Medication history retrieval (with consent)          | 3.8 |     |    |              |          |
| 11 | Multi-location / multi-practice prescribing          | 3.9 |     |    |              |          |

| #  | Feature                                                          | §    | Yes | No | Nice-to-have | Not sure |
|----|------------------------------------------------------------------|------|-----|----|--------------|----------|
| 12 | Multi-tenant / platform deployment (multiple downstream clients) | 3.10 |     |    |              |          |
| 13 | Migration from an existing e-Rx system (Change of Vendor)        | 3.11 |     |    |              |          |

## Section 3: Feature Deep Dives

*Cover each feature flagged Yes or Nice-to-have. Each question below maps to a row in the table beneath it, in order. The goal is a clear recommended approach by the end of each section.*

| Lane                         | Subsections | When to read                                                                |
|------------------------------|-------------|-----------------------------------------------------------------------------|
| <b>Foundations</b>           | 3.1 – 3.3   | Always                                                                      |
| <b>Controlled Substances</b> | 3.4         | Only if you prescribe controlled substances (1.1)                           |
| <b>Prescribing Workflow</b>  | 3.5 – 3.8   | Always                                                                      |
| <b>Practice Structure</b>    | 3.9 – 3.10  | When prescribing across multiple locations, entities, or downstream clients |
| <b>Situational</b>           | 3.11        | When replacing an existing e-Rx system (1.3)                                |

## Foundations

### 3.1 Integration Model

How much of the prescribing UI you build — the decision that shapes almost every section below.

#### Questions:

- Do prescribers need to write prescriptions *in context* (from inside your chart), or is a standalone screen fine?
- How much engineering do you want to invest in the prescribing UI, now and ongoing?
- Do you need to control the look and steps of the flow, or is a vendor-standard flow acceptable?

| Situation                                 | Approach                                                                                                                                                 |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standalone screen fine; minimal build     | <b>Hosted iframe</b> — the vendor's prescribing UI, embedded. Fastest to live; UX is fixed and not customizable.                                         |
| Must prescribe from within your own chart | <b>Integrated (API-driven)</b> — build the pre-approval flow on FHIR operations and bots; the prescriber stays in your UI until the final send.          |
| Own the branding and steps                | <b>Integrated</b> — you control everything up to the vendor-hosted review-and-send widget (stylable; transmitted content is not).                        |
| Either model                              | The final review-and-send step is always vendor-hosted and compliance-controlled — neither model replaces the moment of transmission with your own code. |

 **Choose deliberately.** This is expensive to reverse — it determines whether you build a prescribing UI at all, whether “prescribe from the

chart” is possible, and how enrollment and favorites surface. Align on 1.2 before committing engineering.

### 3.2 Prescriber Enrollment & Identity Proofing

Every prescriber must be enrolled before transmitting. Decide who drives enrollment and how much your platform automates.

#### Questions:

- Who initiates enrollment — an ops admin, or the prescriber on first login?
- Do prescribers send controlled substances (which adds identity proofing and 2FA — see 3.4)?
- How will you see which prescribers are stuck mid-enrollment?

| Situation                            | Approach                                                                                                                                                                                                     |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ops admin drives it, stage by stage  | <b>Admin-driven enrollment bot</b> — invoked per practitioner; the admin controls when each stage advances. Requires a project admin with vendor admin access.                                               |
| Prescriber self-enrolls on first use | <b>Self-service enrollment bot</b> — auto-advances through registration as the prescriber opens the iframe. Requires an admin-level vendor user id as a project secret and an active prescriber role record. |
| Non-controlled only                  | Enrollment completes at basic registration — the prescriber can prescribe non-controlled meds immediately; no identity proofing or 2FA.                                                                      |

| Situation              | Approach                                                                                                                                |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Track enrollment state | The bots return a status that progresses registration → identity proofing → 2FA; surface it to admins so stuck prescribers are visible. |

### 3.3 Provider & Practice Identity Data

Accurate provider identity data prevents the most common enrollment failures.

#### Questions:

- Do you have complete, validated Practitioner records (name, DOB, address, work phone/fax, email, NPI)?
- Do controlled-substance prescribers have a valid DEA number with issuing state?
- Which roles do users need — prescriber, agent/proxy, admin?
- Do non-prescriber staff (front desk, MAs) need to perform prescribing-adjacent tasks — searching pharmacies, setting a preferred pharmacy, viewing history — without an NPI?
- Are any of those staff offshore or otherwise outside the US?

| Situation                       | Approach                                                                                                                                                                                    |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standard prescriber             | Practitioner with full name, birth date, complete address, work phone and fax, email, and a valid 10-digit <b>NPI</b> . Missing or malformed fields are the top cause of enrollment errors. |
| Controlled-substance prescriber | Additionally a <b>DEA number</b> with the issuing <b>state</b> present (2-letter abbreviation) — required to trigger EPCS enrollment (3.4).                                                 |

| Situation                                         | Approach                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Staff who stage but don't send                    | Enroll as a <b>prescribing agent</b> or <b>proxy</b> role — no NPI required; cannot finalize or transmit.                                                                                                                                                                                                                                                                        |
| Users who enroll others                           | Enroll with an <b>admin/clinician-admin</b> role to invite and manage other prescribers and configure favorites (3.7).                                                                                                                                                                                                                                                           |
| Non-prescriber staff running pharmacy/history ops | Even without prescribing rights, staff must be <b>enrolled with a vendor identifier</b> to call pharmacy-directory or medication-history operations — an un-enrolled user is rejected. Confirm which staff need these ops and provision a non-prescribing role for them.                                                                                                         |
| Offshore / non-US staff                           | Prescribing itself is limited to professionals <b>authorized to prescribe in the United States</b> , so non-US users can only take non-prescribing roles (agent/proxy, admin) — and still need vendor enrollment for pharmacy and history operations. Vendor policy on non-US users varies; validate it with your vendor before designing an offshore-staffing model, not after. |

## Controlled Substances

*Read only if 1.1 includes controlled substances. Otherwise skip to 3.5.*

### 3.4 EPCS (Electronic Prescriptions for Controlled Substances)

Supported, but a regulated, multi-step enrollment on top of basic registration. Treat it as its own workstream.

#### Questions:

- Which prescribers will send controlled substances, and does each have a valid, active DEA registration (with state)?
- Are prescribers ready to complete identity proofing and set up two-factor auth?
- Have any prescribers previously enrolled in EPCS with this vendor under another organization?

| Situation                    | Approach                                                                                                                                                                                                                             |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enable EPCS for a prescriber | Three stages on top of registration: <b>(1)</b> registration with DEA present → <b>(2)</b> identity proofing (Experian-based) → <b>(3)</b> two-factor auth activation. All three required before transmitting controlled substances. |
| Missing or inactive DEA      | Blocks identity proofing; the DEA number <b>and</b> issuing state must be on the Practitioner (3.3).                                                                                                                                 |
| Prior enrollment elsewhere   | Can cause verification conflicts (e.g. reused phone number). Identify these prescribers up front so vendor support can resolve them before onboarding.                                                                               |

 **EPCS is a one-way gate on go-live.** Identity proofing and 2FA are per-prescriber, take real calendar time, and are the most common onboarding delay. Start early and build admin visibility into each prescriber's stage (3.2). Controlled-substance prescriptions also require effective dates and often a diagnosis — confirm your workflow captures these.

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## Prescribing Workflow

### 3.5 Formulary/Benefit Checks & Prior Authorization

Whether prescribers see coverage and cost at the point of prescribing, and how prior auth is handled.

#### Questions:

- Do prescribers need formulary status, patient cost, and covered alternatives while prescribing?
- Do you need to initiate electronic prior authorization (ePA) — through the eRx vendor, the standards-based payer path, or both?
- Is your timeline driven by upcoming e-prescribing mandates?

| Situation                        | Approach                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Show coverage & cost             | The vendor pulls active benefit info from pharmacy-benefit data via patient demographics and surfaces formulary status, cost, and alternatives. Coverage comes from the benefit network, <b>not</b> insurance data stored only in Medplum, and prescription cost is <b>not</b> retrievable via the Medplum API — keep demographics accurate so the match succeeds. |
| ePA via the eRx vendor           | The e-prescribing vendor can initiate ePA from the medication when formulary data flags it's required. This may require <b>separate enablement</b> with the vendor rather than being part of the base integration — confirm it's in scope.                                                                                                                         |
| Standards-based ePA (payer path) | The emerging FHIR path for payer-connected prior auth — CDS Hooks with CRD / DTR / PAS (ONC criteria (g)(31)–(g)(33)). Currently <b>alpha</b> .                                                                                                                                                                                                                    |

| Situation                  | Approach                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compliance-driven timeline | HTI-4 (effective Oct 1, 2025) mandates <b>NCPDP SCRIPT v2023011</b> with ePA in the prescribing workflow and <b>RTPB v13</b> for real-time cost display. Payer ePA APIs are due <b>Jan 1, 2027</b> ; eRx modules must be certified to SCRIPT v2023011 by <b>Jan 1, 2028</b> . Factor these into the 3.1 decision. See <a href="#">HTI-4 &amp; CMS-0057-F</a> . |

### 3.6 Refills, Reorders & Discontinuation

How prescribers handle prescriptions after the first send.

#### Questions:

- Do prescribers respond to pharmacy-initiated refill requests?
- Do they re-prescribe previously prescribed medications?
- Do you need to discontinue or expire prescriptions – automatically on change?

| Situation                | Approach                                                                                                                                                     |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refills                  | Pharmacy-initiated requests for more of an existing prescription; prescribers approve or deny.                                                               |
| Reorders                 | Prescriber-initiated new prescription for a previously prescribed medication.                                                                                |
| Discontinuation / expiry | Manual today – no automatic discontinuation-change or auto-expire. High-polypharmacy populations should plan an explicit medication-reconciliation workflow. |

### 3.7 Favorites, Order Sets & Pediatric Prescribing

#### Questions:

- Does the practice repeatedly prescribe a common set of medications worth curating?
- Do you prescribe to pediatric patients?

| Situation                   | Approach                                                                                                                                                                                                       |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Common medication set       | Configure <b>clinic favorites / order sets</b> — curated, reusable entries, including specialty/compounded configurations. Requires an admin/clinician-admin role (3.3); confirm who may create and edit them. |
| Highly variable prescribing | Skip favorites — the maintenance overhead isn't worth it.                                                                                                                                                      |
| Pediatric patients          | The <b>prescriber is responsible</b> for dosages within guidelines for the patient's height and weight; vendors often add pediatric validation — confirm requirements before go-live.                          |

### 3.8 Medication History

Whether prescribers can see a patient's external medication history, and the consent required to retrieve it.

#### Questions:

- Do prescribers need a patient's medication history from outside your system (pharmacy fill history, active prescriptions elsewhere)?
- Can you capture and record the patient consent required to query external history?
- Are you importing prescription history from a prior system?

| Situation               | Approach                                                                                                                                                                               |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Need external history   | Retrieved from the SureScripts network (pharmacy fill history and/or active prescriptions) – a distinct capability from writing prescriptions, so scope it explicitly.                 |
| Consent gate            | External history retrieval requires <b>patient consent</b> ; capture and record it (e.g. a Consent resource) before querying, and confirm the consent language your workflow presents. |
| Importing prior history | History migrated from a prior system generally loses prescriber/pharmacy linkage and may not appear in the vendor’s native “past prescriptions” view – set expectations accordingly.   |

## Practice Structure

### 3.9 Multi-Location / Multi-Practice Prescribing

Read when prescribers work across more than one location or legal entity.

#### Questions:

- Do prescribers prescribe on behalf of more than one practice?
- How should the system know which practice a prescription belongs to?
- Do you need a default pharmacy per patient or per clinic?

| Situation                    | Approach                                                                                                                                                 |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multiple practices/locations | Model each <b>practice</b> as an Organization (linked to a parent business-unit Organization via partOf) and associate prescribers via PractitionerRole. |

| Situation                               | Approach                                                                                                                                                                                                                                                 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resolving the practice per prescription | Precedence: explicit organization id on the operation → the prescriber's single affiliation → a configured default. <b>If a prescriber has multiple affiliations and no organization id is passed, the request errors</b> — pass it explicitly for them. |
| Default pharmacy                        | Settable per patient or per clinic; often irrelevant for nationwide telehealth where patients are everywhere.                                                                                                                                            |

### 3.10 Multi-Tenant / Platform Deployments

Read when one Medplum project serves multiple **downstream clients** that prescribe independently (a platform or enablement model) — distinct from one organization's own locations (3.9). This is an architecture fork worth settling before the second client onboards.

#### Questions:

- Does a single Medplum project serve multiple downstream clients or brands that prescribe independently?
- Must each client's prescribers and patients be isolated from the others?
- Does the originating clinic's identity or branding need to reach the pharmacy?
- Do downstream clients' admins need scoped self-service access to manage their own prescribers?

| Situation                    | Approach                                     |
|------------------------------|----------------------------------------------|
| One organization, many sites | Not multi-tenant — use Multi-Location (3.9). |

| Situation                            | Approach                                                                                                                                                                                                                                           |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Platform serving independent clients | Model each downstream client as its own practice/tenant so prescribers, patients, and rosters are scoped per client rather than pooled.                                                                                                            |
| Roster / patient isolation required  | Scope prescriber pickers and patient lists to the client's tenant — a single shared tenant can expose one client's prescribers or patients to another. Decide isolation before onboarding the second client.                                       |
| Origin branding at the pharmacy      | If the medication must reach the pharmacy carrying the originating clinic's identity/branding, model a distinct tenant per client rather than one shared tenant.                                                                                   |
| Scoped admin access                  | The <b>integrated</b> path can grant client admins access scoped to their own tenant (cross-tenant admin access is blocked for HIPAA); the <b>hosted-iframe</b> path generally cannot expose a scoped admin portal — a factor in the 3.1 decision. |
| Shared vs. per-client tenant         | A single shared tenant is simplest but limits isolation and branding; a tenant-per-client isolates cleanly but adds enrollment and ops overhead (prescribers may need enrolling in each). Choose based on how much the clients must be kept apart. |

## Situational

### 3.11 Migration from an Existing e-Rx System (Change of Vendor)

Read only when replacing an existing EHR or e-Rx vendor (1.3).

#### Questions:

- Are prescribers currently sending through another system on SureScripts?
- Do pending refills and history need to follow prescribers over?
- Can migration be phased prescriber-by-prescriber, or must it be all at once?

| Situation                  | Approach                                                                                                                                                                                             |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Already on SureScripts     | Coordinate a <b>Change of Vendor</b> so the new system is recognized; without it, new prescriptions may work while refills and edits fail.                                                           |
| Preserving pending refills | The approach depends on whether pending refills must be preserved vs. whether the old system must keep operating during cutover — these pull opposite ways. Scope with the vendor before scheduling. |
| Phased rollout             | Migration can generally be coordinated per prescriber, enabling a phased cutover.                                                                                                                    |

## Related Integrations

The e-prescribing vendors Medplum integrates with first-party:

- **ScriptSure** — full API surface with custom FHIR operations and bots, plus an authenticated iframe. The option behind the **integrated (API-driven)** model in 3.1.
- **DoseSpot** — embedded iFrame prescribing, including clinic favorites (3.7) and **prescriber enrollment** (3.2).